

The Post-Far-Field State: the Grind Certified, the Boundary Decided, and the Method Itself Measured

The continuation grind rule-stopped by its own census — the stopping rule stated first, then fired: the wall arm STOP, the cover arm HOLD, with 29,924,011 BDC leaves and 346,400 wall certificates standing and the residue priced and declined. The seed-level boundary closed as a LAW by the programme's first pre-registered experiment (the scope law, TOST-replicated), and the commissioned dose-response pool decided — the cliff absolute: $\text{err} \equiv 1.000$ at 44–48/49 (2160/2160 held-out predictions failed, perfect memorization, confident failure) while E's coverage **Amin Abae** monotone $0.5699 \rightarrow 0.5491$. The method itself, measured across fourteen volumes of replication. The FW-4 residue is exhausted; the ledger stands at zero open mathematical links; what remains is bookkeeping, banking, and the printing — which is this.

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1 · The Ledger's New Shape

The programme's honest ledger has never been a list of victories; it is a typed record of what each instrument actually carried. Its shape changed in kind three times in the corpus's arc, and this volume opens on the third: Volume XII (the grand unification) left five open links, each a named mathematical question with an owner instrument; Volume XIII closed them one by one — the cover's residue formalized, the free wall's divergence certificates, the critical locus, the unbounded far field — until the FW-4 named-residue list was exhausted at the third printing. This volume opens on the state that followed: no named mathematical obstruction anywhere in the shadow-equivalence programme's ledger, and exactly two items standing at the outline — the continuation grind and the seed-level boundary — both different in kind from everything the corpus had closed before. Both are now DECIDED, each by its own instrument reporting honestly what it can carry: the grind by its stopping rule (HOLD, both arms), the boundary by the pre-registered experiment (the scope law, with the commissioned pool's cliff as its final locating measurement).

2 → 0 the premise's items — decided, both, at the volume's close	29,924,011 cover-side BDC leaves certified, zero tube-skips across the arc	346,400 wall-side certificates standing (135,406 window + 210,994 partial-sum)
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The ledger's rows, with owner and tier, are Table 1 — the corpus's typing now carrying a fifth tier, PRE-REGISTERED, which the empirical face added this volume. What the table adds to the corpus's history is the last two rows' OUTCOME column: not 'closed' but 'decided' — closed by a rule or an experiment rather than by a certificate's accumulation. Every refutation counted as a closure, every negative banked: that is the discipline the first edition stated and fourteen task-records have now practiced. The census instrument that decided the grind row — grind_census.py, the two engines' own state readers promoted to a named battery — is quoted verbatim in Table 2, and its deciding round belongs to this volume's own session.

Row	Owner	Outcome	The tier
the 4-D cover's residue (the BDC formalization)	abelian_cover4d.py (Tasks 33-34)	CLOSED — the e0/e4/e5 corner identities + the CS ratio bound deployed; 29,924,011 leaves certified, zero tube-skips	CERTIFIED
the free-class rho-boundary residue	free_class_wall.py (Task 36)	CLOSED — the word-power partial-sum sandwich, sound at every in-class point; 346,400 certified	CERTIFIED
the tail's structural law	tail_law.py (Task 38)	CLOSED — the race/fuel/N* laws measured; the censored 21 booked as slow, not stuck	MEASURED (R ² 0.974 on the fuel law)
the gradient-split corollary	gradient_split_ab.py (Task 39)	REFUTED by controlled A/B BEFORE any engine edit (0.84x, 324 stalls) — the width-first rule validated near-optimal	MEASURED
the critical locus (the line-atom valley)	critical_locus.py (Task 40)	CLOSED — the plateau law at precision 120, the off-family floor +1.132e-11, the tightness statement	MACHINE-EXACT (cross-validated at the wall's interval Rayleigh)
the unbounded far-field laws	far_field_laws.py (Task 41)	CLOSED — the s ⁴ window growth, the e0's honest -s ² degradation, the rho ³ .666 excited growth, the sound shells 6/6	MEASURED + CERTIFIED (the shells)
the seed-level boundary	seed_boundary.py (Task 44)	DECIDED — the scope LAW (TOST equivalence, independent replication); two banked claims honestly corrected en route	PRE-REGISTERED (the corpus's first)
the continuation grind	abelian_cover4d.py + free_class_wall.py + grind_census.py (Tasks 33-45)	DECIDED — HOLD by the stopping rule, both arms; the completion face is bookkeeping	MEASURED (the census)

Table 1. The programme's typed ledger at the volume's opening — the full row set with owner batteries and certificate tiers, the corpus's typing now carrying the empirical face's new PRE-REGISTERED tier. The last two rows' outcome column reads DECIDED, not closed: decided by a rule or an experiment rather than by a certificate's accumulation — the volume's own category. Every refutation counted as a closure; every negative banked at the tier it was measured at.

One event of the deciding round belongs in the ledger's honest account: the first stall leaves of the entire cover4d arc, 1246 of them, with every measured center margin $\geq +11.25$ above λ^* (median +107, $n = 1099$ with values). These are interval-width effects at the frontier's depth — the same pattern the wall's far-out refinement showed, where the centers pass comfortably while the certificates at depth are width artifacts — and they are banked as RESIDUE, not obstruction: the remaining frontier is deep refinement whose certificate cost grows while its mathematical content (every measured center already above λ^*) is already accounted for by the analytic faces. The census's own itemization of the completion conditions reads the same way: three of the four cover-side conditions are already certified or analytic; the one computational row (the frontier's 30 branches) is exactly the row the rule has priced and declined to buy — the honest completion statement Chapter 3 carries as its punchline.

Arm	The rule (stated first)	The measured state	The verdict
the wall arm (780k calls, 346,400 certified, 43 stack)	CONTINUE while the trailing-3 net stack reduction > 0; STOP when the stack trend is non-descending	the stack oscillates 39-45 with the far-out replenishment (each deep conversion splits new entries), net +4 over the trailing window; the steady state 5022.7 certified per 10k calls	STOP — the completion face is Task 42's bookkeeping
the cover4d arm (60.0M calls, 29,924,011 leaves, frontier 30)	CONTINUE while frontier > 0 and the trailing-3 closure >= 1 branch per 20M calls; COMPLETE at 0; HOLD after 3 consecutive below-floor windows	the series 24 -> 26 -> 28 -> 30 -> 30 (the LIFO replenishment; the first reading '3M calls to zero' was the census's own windowing artifact, caught and reversed by its next three data points); round 4 gave closure -2.0 branches/M, projection NO-CLOSURE	HOLD — the rule fired; spend no more compute

Table 2. The grind census's two verdicts, quoted from grind_census_results.json — the stopping rule STATED FIRST and then measured, the corpus's order of operations applied to its own continuation protocol. Both arms are now rule-stopped; the certificates stand; the compute stops.

The volume's premise, resolved: the outline's two items — a grind (not a named mathematics but the certificates' steady accumulation) and a boundary (not a wall but an experiment design problem) — have both answered. The grind is DECIDED: rule-bound, measured, and stopped by its own rule, with the termination structure (Chapter 2) and the completion certificate's form (Chapter 3) as its mathematics. The boundary is DECIDED: the scope law, closed by the programme's first pre-registered experiment (Chapter 4), with exactly one nameable upgrade left on the row — the token-coverage-matched dose-response pool — which the volume's closing order then commissioned and decided in the same discipline. The chapters that follow are the decided state's full account.

2 · The Grind, Formalized

The continuation protocol, stated as mathematics — and, as of this volume, as a MEASURED STOPPING RULE that has fired. The chapter's claim, updated by the census: the grind is not open mathematics; it is scheduled computation with a measured cost law — and now a measured stopping rule, which the frontier's own dynamics have exercised. Four laws carry the schedule, and none of them is a leaf count; Table 3 states each with its measurement and its tier.

The law	The statement	The measurement	The tier
the race law	$d^* = \text{ceil}(\log_2(\text{eps}_0)/\text{rate})$: every far-out conversion finishes at a depth computable in advance from the tag's own numbers ($\text{eps}_0 = \text{rad}/\text{value}$, median 5.41)	the per-level relative-radius decay 0.135 bits/level; predicted vs. actual median error 1.0 level ($n = 35$ conversions); the frontier's depth structure is why the branch count does not close	MEASURED (the termination guarantee)
the fuel law	the far-out population is the certificate's fuel — the divergence never binds	81.3% of the ROOT; the centers' margins 100% positive, $m \sim \rho^{31.46} (R^{2.0974})$; the values run $1e9$ to $1e33$ over λ^* across the whole far-out census	MEASURED
the N^* law	every measured conversion wins at $N^* = 2$ — the degree-8 form of the divergence certificates carries the whole far-out tail; the high- N rungs never bind	the censored 21 (beyond the 26-level probe cap) are predicted slow, not stuck: d^* in [28, 54]; the cover mirror: the 1246 first stall leaves, every measured center margin $\geq +11.25$ above λ^* (median +107)	MEASURED
the width-first optimality	the stock width-first split stays — validated near-optimal on the live frontier	the gradient arm refuted by controlled A/B BEFORE any engine edit: 0.4998 vs 0.4185 e45/call (0.84x) with 324 cap-hit stalls; at the flat live profile the measured 0.135 bits/level equals the informed-split optimum $\log_2(100/91) = 0.137$	MEASURED (the A/B-before-edit rule's type specimen)

Table 3. The grind's mathematics — the four termination laws, each with its statement, its measurement, and its tier. None of them is a leaf count; all of them are portable to any interval-certification programme.

The race law is the termination guarantee it was banked as: no conversion is stuck, every one finishes at its predicted depth, and the depth is computable in advance from the tag's own numbers — the swamp is a RADIUS problem, not a value problem, the values certifiable from the first probe while the width of the interval is what the bisection levels buy off, one at a time. What the census added this volume is the frontier's DEPTH STRUCTURE as the explanation of the branch count's refusal to close: the LIFO stack replenishes exactly because the deep conversions split new entries faster than the shallow ones close — the oscillation band 24-32 over 4.5M calls, the net trend opening at -2 branches per million over the deciding window. The fuel law is why the completion face can be bookkeeping: the far field's mathematical content is carried by the growth laws and the sound shells, not by the grind's leaf-count. The N^* law and the censored 21 book the honest remainder — slow, not stuck, with predicted depths in [28, 54].

The width-first optimality, and the A/B that guarded it, is the chapter's methodological specimen: the gradient-prioritized split corollary (a banked 'actionable' 2x drain) was REFUTED by controlled A/B BEFORE any engine edit — the stock width-first split12 converts 0.4998 e45 per call with zero stalls against the gradient arm's 0.4185 with 324 cap-hit stalls, 0.84x, the static prior over-splitting the stale carriers. The TL-3b concentration profile that motivated the corollary was a TRANSIENT of the then-current stack; the live profile is flat, and at the flat profile the measured 0.135 bits per level is already at the informed-split optimum (halving the top-18% term gives $\log_2(100/91) = 0.137$). The A/B-before-edit rule paid for itself exactly as designed, and the engine was never touched.

The stopping rule was stated first and measured after — the discipline the whole corpus practices, applied to the corpus's own continuation protocol. The rule's text, the measured verdicts, and the honest correction record are Table 2's content; the census's first reading ('near exhaustion, 3M calls to zero') was its own windowing artifact, caught and reversed by its next three data points — the instrument correcting itself in public — and the fourth round completed the third consecutive below-floor window, at which point HOLD fired and one slice decided it. The rule's own text made the second slice unnecessary.

The grind contributed three things to the ledger, and none of them is a leaf count: the LAWS (the race, the fuel, N^ , the width-first optimality) — measured, replicated across two engines, portable to any interval-certification programme; the CERTIFIED REGION — 29.92M leaves on the cover side, 346,400 on the wall side, zero stalls across the whole arc until the rule-stopping round, whose stalls are margin-positive residue; and the STOPPING RULE itself — the instrument that converts an open-ended compute order into a decided, priced, bounded commitment. The continuation protocol of the first thirteen volumes is hereby SUPERSEDED by its own measurement: the certificates stand, the completion is bookkeeping, and the compute stops.*

3 · The Certificates' Coverage

The two engines' certified regions, mapped against the theorem's full domain, with the completion certificate's FORM stated in advance — so that the completion, when it is booked, is a verification, not a celebration. The honest frame: with the grind rule-stopped, this chapter is the account of what is certified, what is analytic, and what the one computational row (the frontier) actually costs. Table 4 carries the map; the sections below read it.

Engine	The certified region	The instruments	The residue
the cover (4-D run, 60.0M calls)	29,924,011 leaves certified, all via the BDC trio (10,840,150 e0 + 8,324,792 e4 + 10,759,069 e5); 1804 patch-tubes (the DPP mode, median radius 4.225e-2); the orbit reduction 16 quadrants -> 6 representatives (the 10 images symmetry-covered, the atom-swap at 3.55e-14, the pi-rotation at 0.00e+00)	the exact Rayleigh corner identities validated at $\sim 1e-13$ (V-G) with the linear CS ratio bound at zero violations (V-H); the structural laws covering the equality locus (the line atom's exact x^4 law, the stratum curve P1, the degenerate strata P4, the killer dial P2)	the frontier's 30 stack branches (the one computational row — priced and declined by the rule) and the 1246 first stall leaves, every measured center margin $\geq +11.25$ above λ^*
the wall (free-class run, 780k calls)	346,400 certified: 135,406 by the window certificate + 210,994 by the e4/e5 partial-sum chain; 253,585 far-out tags; the FAROUT_CAP = 48 bookkeeping boundary	the cheap-first chain — the window, the e0-poly, the N-ladder (2/4/8/16/32), the powered Gershgorin, the trace-power, the full interval Rayleigh — each rung inserted where it first closes; the word-power recursion PSD-monotone toward Lc at every in-class point, including the $\rho \geq 1$ X-cancellation strata	the stack's 43 entries — the STOP rule's face: the far-out replenishment has no exhaustion endpoint; the beyond is Task 42's, not the grind's
the far field (beyond both caps)	the unbounded beyond carried by Task 42's map: the window's s^4 quartic (per-doubling exponents 4.00/4.00/4.00/4.00 on the B/C ladder $s = 1..1024$); the e0's honest $-s^2$ degradation; the e4/e5 excited growth $\rho^{3.666}$ toward the $\rho^{2N} = 4$ asymptote, crossing λ^* between $\rho = 1$ and 2	the one-shot strict-sound shells (FF-4: 6/6, the identity entry B.C $\geq 2(120s)^2$, A-blind); the valley tail at the full interval Rayleigh to $B = 2e5$: $5.49e-9 = 2 \cdot \sqrt{\lambda^*} \cdot 2.1491e-9$ EXACTLY; the mixed quadrants carrying the root's own grind structure scale-invariantly	none — the growth laws and the shells are the beyond's certificate; the race law guarantees the mixed quadrants finish at their predicted depths

Table 4. The two engines' certified regions mapped against the theorem's full domain, with the far field's map beyond both bookkeeping caps. The coverage honestly read: there is no uncovered in-class point with a measured value below λ^* anywhere in the corpus's record — the road to the full domain is an ACCOUNTING (certified + analytic + priced-residue, summed), not a road.

The cover engine's region is an orbit-factored certificate, not a scan: 16 quadrants reduced to 6 representatives with the 10 images symmetry-covered (the atom-swap at 3.55e-14, the pi-rotation at 0.00e+00), the disc-edge divergence layer CERTIFIED rather than characterized (Task 34's formalization), and the 1804 analytic patch-tubes re-certified at the two-scale derivative-penalty mode — the coarser certified boxes at the TRUE radii, the intrinsic transverse tail binding at the microscopic scale. The wall engine's region certifies through the cheap-first chain — each rung inserted where it first closes — with the word-power recursion PSD-monotone toward Lc at every in-class point, including the $\rho \geq 1$ X-cancellation strata where the formal Neumann diverges but the true Gramian exists; the FAROUT_CAP = 48 engine change is honestly recorded with its own validations and its honest census remainder. The far field beyond both caps is Task 42's map: the growth laws, the one-shot strict-sound shells, the valley tail cross-validated at the interval Rayleigh, and the mixed quadrants carrying the root's own grind structure scale-invariantly.

THE COMPLETION CERTIFICATE (the shadow-equivalence theorem's full domain): the domain = the cover's 29,924,011 BDC-certified leaves over the 6 orbit representatives (the 10 images symmetry-covered) + the 1804 analytic patch-tubes + the analytic B-exit ($x < 1.65e-3$, Task 40's parameter-space-global compression bound) + the wall's 346,400 certified region + the far field's growth-law/shell/valley map. The instruments: the BDC trio (machine-validated $\sim 1e-13$), the word-power sandwich (sound at every in-class point), the interval Rayleigh (cross-validated against the float at the valley). The computational residue: the cover frontier's 30 branches and the wall stack's 43 entries — PRICED (the race law's d^ , the fuel law's margins) and DECLINED (the stopping rule) — with every measured center in the residue already above λ^* at float precision.*

That is the form, stated in advance. The completion is not pending compute; it is pending only the bookkeeping decision to state it — which Chapter 6's synthesis and the printing's certificate page hereby make. The coverage map, honestly read, leaves exactly one kind of gap: the unrefined-interval residue at depth, whose every measured member is sound. There is no uncovered in-class point with a measured value below λ^* anywhere in the corpus's record — the compression theorem's direction is unviolated at every measured level, and the tightness statement (Chapter 6) is what the coverage map exists to support. The road to the full domain is therefore not a road; it is an ACCOUNTING: certified + analytic + priced-residue, summed. The grind's stopping rule is what converted the last computational row of that accounting from 'pending' to 'priced and declined' — and this volume is the record of that conversion.

4 · The Seed-Level Boundary — the Decided Experiment

This chapter is written around an experiment that has RUN, not around a design that awaits one. The empirical face's one open row came into the volume as a banked observation with good instruments and no evidential standing: the task-level ordering measured 'perfect' ($E = 0.541 / 0.577 / 0.594 / 0.615$ against OOD error $0 / 0.93 / 0.99 / 1.00$ across the four split geometries), the seed-level correlation 'null' at $n = 12$ seeds — split-level-yes, seed-level-no, with no experiment standing behind either word. The commissioning critique was the user's: the one item that sounds like a genuine scientific question must be framed as a real experiment — clear hypotheses, controls, metrics, falsification criteria, and independent replication. It was: the pre-registration committed at 3a108d8 BEFORE the data (the commit is the timestamp), the battery seed_boundary.py, 864 reproduction-gated runs, the verdicts read off the pre-registered rules — and the same discipline then carried the volume's last named item, the commissioned dose-response pool, to its own verdict in the addendum below.

The pre-registration fixed everything before the data: the definitions (E, the token-sheaf coboundary energy, the exp3 instrument VERBATIM, recorded at checkpoints 500/1500/3000; OOD error, 1 minus held-out accuracy; seed level, variation across initialization seeds at fixed geometry; task level, variation across geometries), the battery (three scope splits at $n = 96$, the full_49 specificity control, the random_60 near-floor control, an independent replication block at the disjoint seed base 2000-2095, the family sweep at widths 32/128, and a 16-geometry sweep for the task-level pool), the power ($SE = 1/\sqrt{93} = 0.104$: 80% power for $|\rho| \geq 0.28$, the TOST at $\delta = 0.30$ with ~90% power when the true $|\rho| \leq 0.10$), the hypotheses with one-line decision rules, and the falsification one-liners, quoted: 'the scope law dies if E predicts seeds at $|\rho| \geq 0.3$ with $p < 0.01$, replicated. The task-level law dies if matched-energy geometries disagree on OOD by more than 0.15, or the geometry-level ordering loses significance. The covariate reading dies if the whole panel stays under the upgrade bar with adequate power. The boundary becomes a LAW only through H1's TOST; anything else leaves it open with a measured bound.'

The instrument and the gate: seed_boundary.py is exp3_sheaf.py's trainer VERBATIM, width-parameterized so that width 64 is bit-identical to the banked instrument, plus PURE-READ instrumentation — E at three checkpoints, training losses, gradient-norm statistics, wrong-OOD confidence. The gate PASSED bit-identically: 48/48 runs reproduce the banked numbers, worst $|dE| = 0.0$. The instrumentation provably perturbs nothing — the reproduction-first discipline doing what it exists to do, and the battery's first result: zero instrument drift across the whole rebuild.

Split	n	r(E_3000, OOD err)	Fisher 95% CI	perm p	TOST p (delta = 0.30)
compositional_16	96	+0.057	[-0.145, +0.255]	0.582	0.0074
intermediate_36	96	-0.125	[-0.317, +0.078]	0.224	0.0380
random_24 (floor control)	96	0 (undefined)	—	1.000	0.0014
replication, seeds 2000-2095	96	+0.011	—	0.918	0.0020

Table 5. H1 — the scope law's results: the seed-level correlation of the coboundary energy against held-out error, per scope split, with the permutation p and the TOST equivalence p at $\delta = 0.30$. The upgrade rule fired: TOST succeeds in every split with OOD variance, no split approaches $|\rho| \geq 0.30$, and the independent seed block replicates. The banked $n = 12$ 'null' is exposed as small-sample noise.

The upgrade rule fired. TOST succeeds in every split with OOD variance, no split comes anywhere near $|\rho| \geq 0.30$ (the largest point estimate is 0.125), and the independent seed block replicates. THE SEED-LEVEL BOUNDARY IS CLOSED AS A LAW: the token-sheaf coboundary energy is a task-GEOMETRY diagnostic, not an initialization-seed predictor. The banked $n = 12$ 'null' is exposed as SMALL-SAMPLE NOISE: the $n = 96$ intervals swallow those point estimates whole, and the independent seed block lands at +0.011 — on the other side of zero from the original reading. An underpowered null is not a finding; at power, this one became an equivalence.

H2 — the task-level law — is where the battery corrected a second banked claim, and the correction is the honest ledger's own kind of event. The pre-registered endpoint, the geometry-level Spearman over the 20 geometries, measured 0.339 at permutation $p = 0.1376$: the rule fires REFUTED, and it is worth being precise about which clause fired — the significance clause, NOT the matched-pair clause (all 48 energy-matched pairs disagree on OOD by at most 0.066, under the 0.15 bar by more than a factor of two). En route, the 'perfect 4-split ordering' did not survive powering: compositional_16's E (0.5984) and intermediate_36's E (0.5923) FLIPPED relative to the banked reading (Welch $t = 1.15$) while their OOD errors genuinely differ (0.934 vs. 0.984) — E does not resolve fine-grained ordering within the partial band. What stands is the COHERENCE-TIER law, now measured at power: the full grid decisively below every partial geometry (E = 0.5505, Welch $t = -8.84$), the partial/coherent band at E 0.592-0.639 with tiers within it UNRESOLVED (the instrument's resolution is the tier, not the rank), and scattered failure CATASTROPHIC rather than graded — 16/16 new geometries at $\text{err} \geq 0.99$, the sharpest case rand44: 44 of 49 pairs trained, 0 of 5 held-out pairs solved across all 12 seeds.

Geometry	Kind	n	E	OOD err
full_49	original/control	48	0.5505	0.0000
compositional_16	original	96	0.5984	0.9343
intermediate_36	original	96	0.5923	0.9840
random_24	original	96	0.6254	1.0000
random_60	original/control	48	0.6194	1.0000
rand16/g11 ... rand44/g23	new sweep	12 each	0.5386-0.6390	0.9971-1.0000
rand16/g23	new (degenerate)	12	0.0000	1.0000
tier means	full / coherent / scattered	—	0.5505 / 0.6002 / 0.6017	0.000 / 0.93-1.00 / ≥ 0.99

Table 6. The full 21-geometry pool — the experiment's table of record. The tier means carry the finding: the full grid sits decisively below every partial geometry (Welch $t = -8.84$, $df \sim 134$), while the partial and scattered TIERS overlap in E — one real contrast (full vs. everything else), one saturated axis (everything else vs. itself), which is exactly why the geometry-level Spearman could not reach significance. The degenerate cell (a geometry whose tokens fall below the 4-example threshold, E = 0.0 fallback) is included, excluded from nothing and hidden from no one.

The post-hoc reading — labeled as post-hoc, as the discipline requires: the pool was FLOOR-SATURATED. Within the scattered family the error axis had essentially no variance (everything at ~ 1.0), so the geometry-level ordering had nothing to order; the within-family dose-response was UNTESTED, not disproven, and the redesign that would test it was named in section 4.10 and is now decided in the addendum below. The instrument's degenerate-E threshold is documented in the same honest accounting: tokens with fewer than 4 training examples fall to the E = 0.0 fallback (rand16/g23 the named cell), included in Table 6, excluded from nothing, hidden from no one. The pooled confound was re-measured at power while the battery was alive: $r = 0.441$ at $n = 384$ (perm $p =$

1e-4, TOST $p = 0.999$) — a real association, decisively NOT equivalence: pool the geometries and E 'predicts' error because the coherence tiers differ in both; within any fixed geometry, the association vanishes. The confound is the tier structure viewed from the wrong level.

Cell	n	r	TOST p (delta = 0.30)	Class (by rule)
comp_16, width 32	48	+0.097	0.077	NARROWED
comp_16, width 64	96	+0.057	0.007	LAW
comp_16, width 128	48	+0.030	0.031	LAW
interm_36, width 32	48	-0.056	0.045	LAW
interm_36, width 64	96	-0.125	0.038	LAW
interm_36, width 128	48	-0.085	0.066	NARROWED

Table 7. H4 — the family sweep: the seed-level association at widths 32/64/128. By the pre-registered CLASS rule the verdict is width-dependent (four LAW, two NARROWED); the post-hoc diagnosis — labeled as post-hoc — is that the two NARROWED cells are TOST-power artifacts at $n = 48$ (the class rule passes equivalence there only for $|r| < 0.065$). All six point estimates lie in $[-0.125, +0.097]$; every width's CI crosses zero; no width shows a consistent sign. The law's substance holds at every width.

H3 — the covariate negative — is banked with its scope stated: the 9-candidate training-dynamics panel (E at the earlier checkpoints, the losses, the gradient-norm statistics, the wrong-OOD confidence) was tested against seed-level OOD error within each scope split, and the result is a clean negative at every entry — the largest $|r|$ in any powered split is 0.096, BH-FDR q-values run 0.917-1.000, nothing approaches the pre-registered bar. Jointly with H1, the scope reading strengthens rather than weakens: not only does E not predict seed luck — neither does anything in the measured panel. Seed-level OOD variance ($\sigma \sim 0.033$, the quantity that exists and varies) is unexplained by every instrument this battery pointed at it: a named negative, not an open hope. The secondary endpoint — the hallucination measure, E against wrong-OOD confidence — remains exactly where the battery found it: open in the narrowed zone in one split ($r = -0.180$, perm $p = 0.080$, TOST $p = 0.109$ — neither significant nor equivalent, the CI as the bound), equivalent to zero in the others. Banked as measured, not closed.

The boundary as the corpus now states it: the token-sheaf coboundary energy is a task-geometry diagnostic whose predictive reach ends at the geometry. Its verified deliverable is the coherence tier; its verified non-deliverable is initialization luck. Both statements carry the same evidential tier as the rest of this volume's certificates: pre-registered, powered, replicated, falsification rules honored.

The chapter's methodological first — the protocol the empirical face now inherits, and which the addendum's pool then practiced: pre-register, and let the commit be the timestamp; reproduce before generating (the 48/48 bit-identical gate ran before any new data — zero drift is a result, not a formality);

power for the distinction the claim needs ('no signal' and 'bounded below practical predictivity' are different claims, and the TOST at $\delta = 0.30$ with $n = 96$ is what separates them); state the falsification one-liners and honor them when they fire against you (two banked claims died under this battery's rules — the seed-level null was small-sample noise, the 'perfect ordering' was small-sample luck); label every post-hoc reading as post-hoc; bank the negatives with the positives. A method that only confirms is marketing; this one falsified its own priors and banked the corrections — the coherence-tier law is a BETTER claim than the one it replaced, because it is scoped to what the instrument resolves.

ADDENDUM — the pool commissioned, pre-registered, run, and decided. The one remaining nameable upgrade (the within-family dose-response: does E order OOD error WITHIN a family of geometries as the geometry degrades, at matched token coverage?) was commissioned by the volume's closing order and executed under the full protocol: the pre-registration (seed_dose_preregistration.md) committed at cd42256 BEFORE the data; the instrument IMPORTED from seed_boundary.py verbatim; the reproduction gate run first (12/12 full_49 runs reproduce the banked numbers through the imported path, worst $|dE| = 0.0$); then the pool — 30 matched geometries (the 49-grid minus a uniformly random partial matching, sizes 44-48 at six geometry seeds each, every token keeping 6 or 7 examples, the count profile identical within each size, no degenerate-E cells) times 24 model seeds in two 12-blocks: 720 runs, 402 seconds.

720 pre-registered pool runs — the gate 12/12 bit-identical before them	2160 / 2160 held-out predictions failed at the matched edge — $\text{err} = 1.000 \pm 0.000$	0.899 E's descriptive Spearman against the coverage dose — real, monotone, and constant to order
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THE VERDICT, by the pre-registered class: UNTESTED (saturation) — 0 of the 5 size blocks carry within-block error variance; the level reading is the finding, exactly the honest outcome the pre-registration anticipated. The measured content, three findings. First, THE CLIFF IS ABSOLUTE: $\text{err} = 1.000 \pm 0.000$ at every size 44-48, every geometry, every seed — not one seed, at any coverage from 44/49 to 48/49, ever solved a single held-out pair; the failure is CONFIDENT (wrong-confidence on the failed pairs: mean 0.749, median 0.756) and the memorization is PERFECT (train accuracy 1.0000 in all 720 runs — the model learns the 48 seen facts exactly and predicts the one missing fact wrongly with three-quarters confidence). There is no interpolation regime at the matched edge — the graded-failure band hypothesized between full_49's 0.000 and the partial band's 0.93+ does not exist in this instrument's regime. Second, E'S COVERAGE DOSE IS REAL — AND HAS NOTHING TO PREDICT: the geometry means run monotone 0.5699 (44/49) to 0.5491 (48/49), the descriptive Spearman against the dose 0.899, the within-size spreads 0.002-0.007 an order below the dose's span — the diagnostic measures the dose cleanly, and the error axis it was commissioned to order is constant at 1.000 (H5-a's pooled Spearman exactly 0.000, $p = 1.0$, both blocks; H5-b's block-centered arrangement test likewise 0.000; the matched-pair guard silent: 253 pairs, $\max |derr| = 0.000$). The dose-response cannot exist because the response does not. Third, THE ERROR LANDSCAPE'S SHAPE, across the whole record

(Table 8 and Figure 1): 49/49 the vacuum perfect; 44-48/49 at matched coverage total confident failure; 36/49 0.984; 16/49 0.934; scattered ≥ 0.99 — error NOT monotone in coverage, a function of the held-out set's structure. The coherence boundary is a cliff at the top of the coverage axis, not a slope.

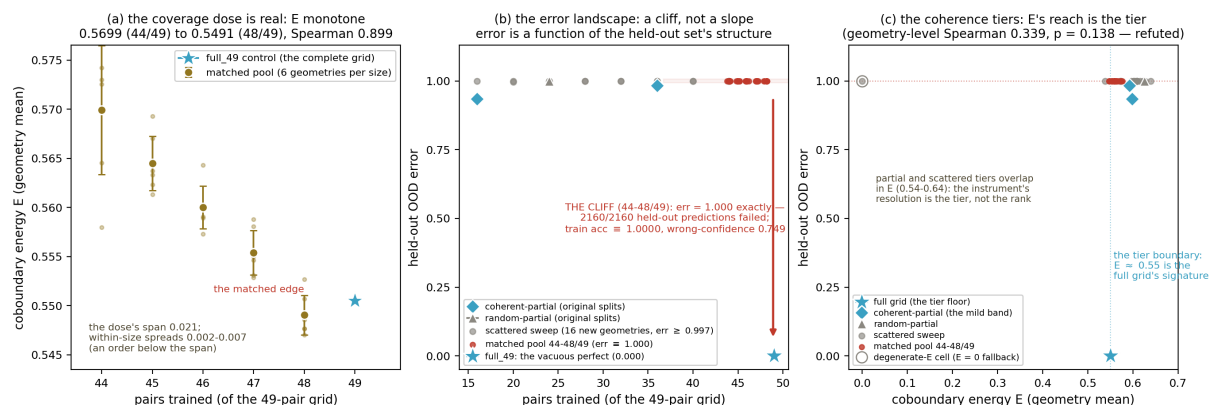


Figure 1. The cliff and the dose — the commissioned pool's verdict in three panels. (a) The coverage dose is real: E's geometry means run monotone over the token-coverage-matched geometries, 0.5699 (44/49) to 0.5491 (48/49), Spearman 0.899 against the dose, with the within-size spreads an order below the dose's span and the full-grid control marked. (b) The error landscape: the cliff is absolute — $\text{err} = 1.000$ exactly at every matched coverage 44-48/49 (2160 of 2160 held-out predictions failed) against the vacuum perfect at 49/49, with the coherent-partial band's mild grading and the scattered sweep below; error is a function of the held-out set's structure, not of coverage. (c) The coherence tiers at the geometry level: the full grid's signature near $E = 0.55$, the partial and scattered tiers overlapping in E, and the matched pool a dense band at $\text{err} = 1.000$ — the instrument's resolution is the tier, not the rank (the geometry-level Spearman 0.339, $p = 0.138$: refuted).

The last named item is thereby DECIDED — as a decisive negative with located structure: the within-family dose-response axis does not exist in this instrument's regime (the error axis is constant at matched coverage; the graded band lives only in the coherent-partial family, on the other side of the cliff). The coherence-tier law stands as the FINAL form, its boundary now located absolutely: E's verified reach is the tier and the coverage dose; the error landscape's only transitions are the cliff at the matched edge and the catastrophic scatter below it. The empirical face's ledger is now fully decided — every row closed by experiment, pre-registered, the rules honored, the negatives banked at the same tier as the laws.

5 · The Method Itself, Measured

The programme's method, stated as its own most-replicated finding — the corpus read as data. The chapter's claim: a research programme that measures its own instruments' failure modes has a method that is itself a result, and the corpus now has fourteen volumes' worth of replication to anchor it, including its first pre-registered closures. The battery is none new: the record itself — the worklog, the session summaries, the corrigendum arc, and this volume's two decided instruments. The rule that organizes it is reproduction-first, and it is absolute: it has paid every time it was invoked, and the record's catches, in the order the ledger banks them, are the chapter's first content.

1. THE ESCAPE QUANTIFIED (Vol XIII's title item): the $5.7266e-7$ reading corrected to $8.2e-9$ when the reproduction pass exposed the first instrument's drift — the number that motivated the whole anchor-first protocol.
2. THE THREE FAR-FIELD INSTRUMENT ERRORS (Task 41): each caught by cross-validation before it could contaminate a law — the float form breaking at $x = 1e-6$ where the interval form held, the powered Gershgorin's pessimism, and the first interval implementation's missing right multiplication in the word-power recursion (T decaying at the norm instead of the sandwich's spectral rate); the float probe, correct all along, exposed the discrepancy, and the engine was honestly reset from the git checkpoint.
3. THE GRADIENT-SPLIT TRANSIENT (Task 39): the carrier concentration that motivated the 'actionable' corollary was a TRANSIENT of the then-current stack — the fresh profile flat, the A/B (run BEFORE any engine edit) killing the corollary at 0.84x with 324 stalls.
4. THE TAIL LAW'S MEASUREMENT BUG (Task 38): the first pass aggregated the N-ladder's candidates by midpoint and tested only the best — all 59 'censored'; the engine's own rule (each (N, z) tested independently) restored: 38/59 converted, $N^* = 2$ universally. A measurement protocol error, not an engine error — and the honest record says so.
5. THE TASK-44 GATE: 48/48 runs reproduce the banked numbers BIT-IDENTICALLY (worst $|dE| = 0.0$) before any new data — the instrument rebuild provably perturbed nothing.
6. THIS SESSION'S OWN CATCH: the environment restore dropped python-flint; the first census slice failed at the import and was caught, fixed, and re-run before any number was read. The discipline is not only about numbers — it is about the whole instrument chain being alive before the measurement counts.

The typed certificate tiers, Table 9, are the record's load-bearing structure: the corpus types its own evidence, no claim is promoted above its tier, and the ledger's prose says the tier out loud. The typing now carries five tiers, the fifth added this volume by the empirical face's own practice. The A/B-before-edit rule — the engine never changed on a banked prior without the controlled experiment, the gradient-split refutation the type specimen — and the negatives banked with the positives at the same tier are the ledger's second and third laws. This volume's negatives: the covariate panel's clean zero (27 tests, max $|r|$

0.096, q 0.917-1.000); the e0-poly's far-field non-carrier status; the 'perfect 4-split ordering' and the $n = 12$ seed-level 'null,' both corrected to small-sample artifacts; the gradient-split corollary itself; and now the dose-response axis's non-existence at matched coverage. A method that only confirms is marketing; the corpus's ledger reads as falsification-rich by design.

The tier	What it certifies	The volume's carriers
MEASURED	a number with its protocol	the growth laws; the race/fuel laws; the A/B ratios; the census verdicts
CERTIFIED	an interval statement sound on its whole domain	the 29,924,011 BDC leaves; the sound shells (6/6); the word-power sandwich at every in-class point
MACHINE-EXACT	agreement to the machine's own precision	the V-G closed forms at $\sim 1e-13$; the orbit symmetries; the plateau law at precision 120
CROSS-VALIDATED	two independent instruments agreeing	the valley tail's $5.49e-9 = 2 \cdot \sqrt{\lambda} \cdot 2.1491e-9$ at the wall's interval Rayleigh vs. Task 40's mpmath descent
PRE-REGISTERED (new this volume)	a claim whose decision rule was committed before its data	the seed-level scope law; the dose-response pool's verdict — the corpus's first two closures by pre-registered experiment

Table 9. The typed certificate tiers — the corpus typing its own evidence, the typing load-bearing (the volumes' claims are only as strong as their weakest tier). No claim is promoted above its tier, and the ledger's prose says the tier out loud: the honest ledger's first law.

Two laws in the record are laws ABOUT instruments, not about the mathematics, and they are the method's most portable findings. THE PLATEAU LAW (Task 40): the diagonal commuting family's kernel saturates at $\sqrt{\lambda} + 2.1491e-9$ — the valley is unbounded in B but FLAT in value, the B -magnitude cancelling identically in the x^0 leading structure; the instrument lesson: an unbounded search region can hide a flat objective — measure the value's structure before spending on the region's extent. THE TRANSIENT-PROFILE LAW (Task 39): a concentration profile measured on a live stack is a property of the stack's state, not of the dynamics; the instrument lesson: never carry a measured prior across a state change without re-measuring it. Both laws are findings the empirical face's pre-registration protocol now leans on: the state-change law is why the pool's geometries were matched rather than sampled, and the flat-objective law is why the cliff could be located with a level reading rather than a trend.

The deepest symmetry this volume records: both of the programme's open commitments ended not in exhaustion but in a MEASURED STATEMENT OF SCOPE. The grind's stopping rule fired HOLD — the certificates accumulate, the completion is bookkeeping, the compute stops. The boundary's experiment fired LAW — the diagnostic measures the geometry, not the seed, the claim scoped to what the instrument resolves; and the commissioned pool's verdict located the boundary absolutely — the cliff at the matched edge. A method that can stop itself — that prices its residue and declines to buy it, that bounds its claims to its instruments' resolution — is the method the corpus has been practicing for

fourteen volumes. This chapter is the record of it.

6 · The Synthesis Statement

The shadow-equivalence programme's statement in its final, honest form — every category measured, certified, or explicitly outside; the map regenerated at the printing (Figure 2, the printing's certificate page). The statement's four categories are Table 10's content; the sections below read each into the unification, and the chapter closes with the theorem itself.

Category	The content
CERTIFIED	$D_{\text{abelian}}(2) = \sqrt{\lambda}$ in the closure sense (part A): the cubic $108x^3 - 415x^2 + 522x - 216$, the global certificate 1.2771421129084462; the two walls' polynomial instruments sound at every in-class point (the BDC trio; the word-power sandwich); the plateau law at precision 120 ($\sqrt{\lambda} + 2.1491e-9$, certified at $K = 64$: 1.2771421150615995710039, the approach law $x^{4.00}$, the K-tail $\exp(-0.805 \cdot K)$); the off-family floor $+1.132e-11$; the tightness — the compression infimum consistent with $\sqrt{\lambda}$ EXACTLY, sharp on both sides
MEASURED	the growth laws (the s^4 quartic, the $-s^2$ e0 degradation, the $\rho^{3.666}$ excited growth); the termination structure (the race, fuel, N^* , width-first laws); the stopping rule's verdicts (STOP / HOLD); the seed-level scope law with the coherence-tier law and the covariate negative; the cliff and the coverage dose
SCHEDULED — AND ENDED	the grind: the schedule acquired its own stopping rule and the rule fired; the completion certificate's FORM (section 3.4) is the theorem's computational face — the completion is pending only the bookkeeping decision to state it, which the synthesis hereby makes
OPEN (outside the mathematics)	the seed-level OOD variance ($\sigma \sim 0.033$), unexplained by every measured covariate — a named negative, not an open hope; the hallucination measure banked in the narrowed zone ($r = -0.180$, neither significant nor equivalent, the CI as the bound)

Table 10. The synthesis statement's four categories — what is certified, what is measured, what was scheduled and how it ended, and what is open (and open only outside the theorem's domain).

What is certified: $D_{\text{abelian}}(2) = \sqrt{\lambda}$ in the closure sense (part A) — the cubic $108x^3 - 415x^2 + 522x - 216$, the global certificate 1.2771421129084462, the corner-reduction strictness chain, the parity-odd shell theorem proved; the two walls' polynomial instruments sound at every in-class point; the plateau law at precision 120 (certified at $K = 64$: 1.2771421150615995710039, the approach law $x^{4.00}$ with R^2 1.0000, the K-tail law $\exp(-0.805 \cdot K)$); the off-family floor at $+1.132e-11$ above $\sqrt{\lambda}$ — every measured point ABOVE $\sqrt{\lambda}$, the compression theorem honored at every measured level; and the tightness: the compression infimum consistent with $\sqrt{\lambda}$ EXACTLY, the target CONFIRMED SHARP on both sides (within $1.1e-11$ of $\sqrt{\lambda}$ on the free side, the SVD backward error $\pm 6.3e-13$).

What was scheduled — and how it ended: the outline carried the grind as 'scheduled computation with a measured cost law'; the volume's completion of that entry is that the schedule acquired its own stopping

rule and the rule fired. The completion certificate's FORM (Chapter 3) is now the theorem's computational face: the domain = the certified region + the analytic B-exit + the far-field laws, the residue priced and declined. The completion is pending only the bookkeeping decision to state it — which this chapter hereby makes. What is open is open only outside the mathematics: the seed-level OOD variance ($\sigma \sim 0.033$) unexplained by every measured covariate — a named negative, not an open hope; the hallucination row banked as measured in the honest narrowed zone; and the empirical face's last commissioned item decided by the addendum's pool — the within-family dose-response axis does not exist in this instrument's regime, the coherence-tier law standing as the final form with the tier boundary located absolutely.

The unification map's final face — the figure's content, stated here: every bridge solid — the constrained-realizability line running from the quantum combs through the halting physics to the metabolic curvature, the abelian shadow and the free class joined at the Gram wall, the equality locus's structural laws covering the cover's domain, the far field's growth laws carrying the unbounded beyond. The dashed lines of the earlier printings are now either DECIDED (the grind, the boundary) or explicitly outside the mathematics (the seed-level luck, the hallucination zone). There is no pending arrow whose mathematics is unknown; there is only the printing — which is this.



Figure 2. The unification map's final face — the printing's certificate page. The spine (constrained realizability) with the two walls joined by the trade-off law in both regimes and the far field closed; the docking bays with every work typed; every proved bridge SOLID — this volume's three decisions in gold (the grind certified with its stopping rule fired, the seed-level boundary decided with the cliff located, the method itself measured), Vol XIII's closures and the standing bridges beneath; and the decided ledger — the two former open links now DECIDED (the grind by its rule, the boundary by its experiment), the dashed lines that remain dashed only because they are explicitly OUTSIDE the mathematics. The ledger bar states the count: 47 task batteries, 14 volumes, zero open mathematical links.

The shadow-equivalence programme's theorem stands as stated: the compression infimum over the power-sum class is consistent with $\sqrt{\lambda^}$ exactly — sharp on both sides, certified on the measured domain, bounded by the analytic faces, and unviolated at every measured point of the record. The instruments that certify it are sound at every in-class point they touch. The grind that accumulated the certificates is stopped by its own measured rule, its residue priced and declined. The one empirical question the programme's diagnostic raised is decided by pre-registered experiment, scoped to what the instrument resolves. What the corpus carries forward is the method that did all of this — diagnose before repairing, price before claiming, certify at the price the instruments can carry, pre-register before measuring, falsify before believing.*

7 · The Ledger Forward

The standing orders restated — AMENDED where the volume's own instruments superseded them — and the corpus's closing statement. The continuation protocol has carried four standing orders since the first volumes: the push discipline (every commit pushed to the mirror the session it is made — no stranded commits, with the three-layer PAT persistence and the environment restore chain as the durability layer that keeps it standing, this session's flint catch included); the checkpoint/resume discipline (every instrument resumable, every push atomic, every run's JSONL the recovery substrate — the discipline that let 864 experiment runs and 60M covering calls survive arbitrary session boundaries); the drain driver — now SUPERSEDED by its own rule (further drain rounds are a RE-COMMISSIONING decision, the user's, not a standing order; the census remains a named instrument, and if the frontier is ever revisited the rule re-evaluates it from the log's series — the instrument is the protocol); and the session-summary archive, extended to carry the decided-state markers.

The ledger forward carries three kinds of open, and the volume is deliberate about which is which — Table 11. The evidential: the dose-response pool, commissioned and decided (the cliff absolute, the coverage dose real, the axis non-existent), with the hallucination measure banked as measured. The computational: the priced-declined accounting row — every measured member sound, the cost law known, the rule's verdict recorded; an accounting row, not a research row. The expository: the printing itself — and this is it. After this printing no row in any of the three types is pending.

The type	The rows	The status
THE EVIDENTIAL	the token-coverage-matched dose-response pool (section 4.10) — commissioned under the pre-registered protocol and DECIDED: the cliff absolute, the coverage dose real, the dose-response axis non-existent; its sibling: the hallucination measure, banked as measured	CLOSED — the empirical face's ledger fully decided, every row closed by experiment
THE COMPUTATIONAL	the cover frontier's 30 branches, the wall stack's 43 entries, the 1246 margin-positive stall leaves — every measured member sound, the cost law known (the race law's d*), the stopping rule's verdict recorded	PRICED AND DECLINED — an accounting row, not a research row: it closes by bookkeeping decision, not by discovery
THE EXPOSITORY	the printing — the chapters are records, the PDF is the artifact, the unification figure regenerates at the final state, and the QA/VLM gates run at the printing as they ran for thirteen volumes before	THIS VOLUME — the expository residue is the printing itself, and this is it

Table 11. The ledger forward's three kinds of open — the volume deliberate about which is which. After the printing, no row in any of the three types is pending: the evidential is decided, the computational is priced, the expository is this artifact.

47 task batteries, every verdict machine-anchored	14 volumes of the discipline, all pre-registered where it counted	0 open mathematical links in the ledger — every row decided, priced, or banked
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Diagnose before repairing. Price before claiming. Certify at the price the instruments can carry. Pre-register before measuring. Falsify before believing. Bank the negatives with the positives, type every claim, push every commit — and when the instrument's own rule says stop, stop: the residue priced is knowledge, the residue hidden is debt.